

SENSORS

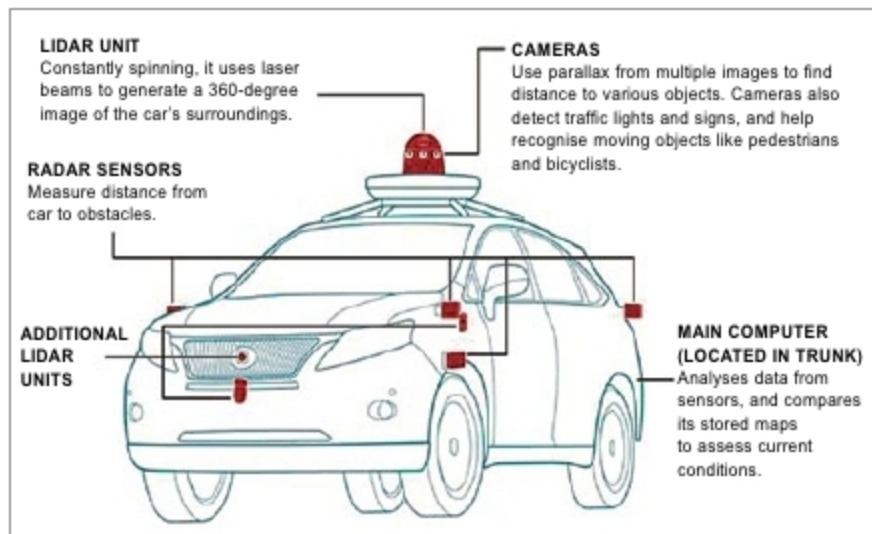


Fig. 4: Sensors used in autonomous vehicles (Credit: www.nytimes.com)

Some leading manufacturers of sensors for automobiles

- Bosch
- Continental
- Delphi Automotive
- Denso
- First Sensor AG
- Honeywell
- Lumina Technologies Inc.
- Micro-Epsilon
- Sensata Technologies
- Wells Vehicle Electronics

tations in current sensor technologies.

Some sensors used in advanced driver assistance systems (ADASes) include fuel delivery, lane departure warning, parking aid, tank pressure measurement, adaptive cruise control (ACC), blind-spot detection, brake booster system, collision

avoidance system, filter monitoring, lidar, power-assisted steering, reversing aid, start-stop system, tank air intake and extraction, tank leakage diagnostics, traffic sign recognition and so on.

To sum up

Today, electronic sensor systems and engine computers do everything, from regulating and monitoring fuel to diagnosing problems. Modern electronic sensors constantly monitor vital engine parameters like oil pressure, coolant temperature and emissions, and report back to the driver when something goes wrong.

Sensors play an important role in the automobile manufacturing industry. Hybrid vehicles require more sophistication in sensing and control. This requires a wide range of sensors, all operating properly. Such dependence on car engine sensors and electronic systems requires a high level of quality and reliability.

Large arrays of sensors are required in modern autonomous vehicles. While there has been good progress in this field, there are still many hurdles to cross. Safety and reliability are still a concern due to limitations in current sensors technologies.

Advancements in smart radar sensors such as lidar and implementation of AI algorithms could help automakers make a truly reliable autonomous vehicle in the near future. **EFY**

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